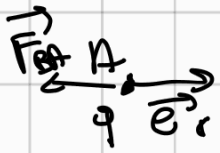
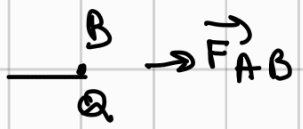
Sayfa 17 Problem 1



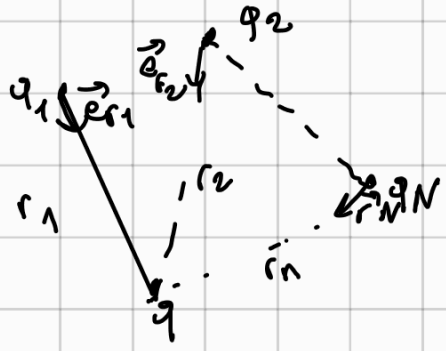


$$\vec{e}_r = \frac{\vec{r}}{r} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$\vec{F}_{AB} = \frac{1}{4\pi\varepsilon_0} \frac{q \cdot Q}{r^2} \vec{e}_r \frac{\vec{r}}{r} = -\vec{F}_{BA}$$

$$\vec{F}_{AB} = \frac{1}{4\pi\varepsilon_0} \frac{q \cdot Q}{r^3} \vec{r} = -\vec{F}_{BA}$$

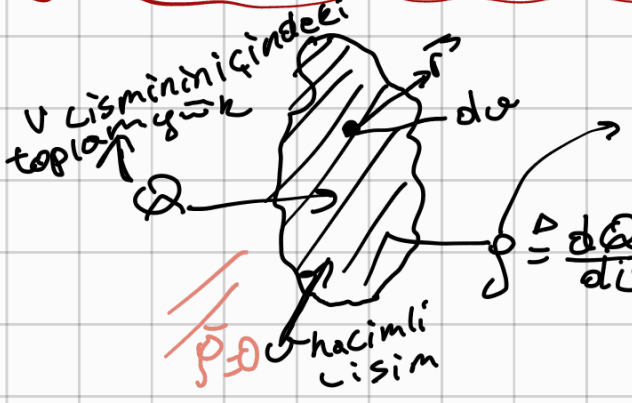
N+1 yük olmadurumu



$$\vec{r}_n = r_n \cdot \vec{e}_n$$

$$\vec{F}_q = \frac{q}{4\pi\varepsilon_0} \sum_{n=1}^{N} \frac{q_n}{(r_n)^2} \vec{e}_n = \frac{q}{4\pi\varepsilon_0} \sum_{n=1}^{N} \frac{q_n}{(r_n)^3} \vec{r}_n$$

Yükler ayrık değilde bir bölgeye dağılmışsa



hacimsel yük yoğunluğu $[\rho] = \frac{C}{m^3}$

$\rho\, dv = dQ$

dv hacim elemanının içindeki yük

$\tilde{\rho} = \bar{\rho} + \rho$

$$d\vec{F}_q = \frac{1}{4\pi\varepsilon_0} \frac{q \cdot dQ}{r^3} \cdot r$$

$$\vec{F}_q = \int_{\vartheta} d\vec{F}_q = \frac{q}{4\pi\varepsilon_0} \int_{v} \frac{dQ}{r^3} \vec{r}$$

$$= \frac{q}{4\pi\varepsilon_0} \int_{\vartheta} \frac{\rho}{r^3} \cdot \vec{r}\, dv$$

$$\equiv \frac{q}{4\pi\varepsilon_0}\int_x\int_y\int_z \frac{\rho}{r^3}\vec{r}\,\underbrace{dx\,dy\,dz}_{d\tau} \triangleq q\cdot\underbrace{\vec{E}}_{\text{elektrostatik alan vektörü}}$$

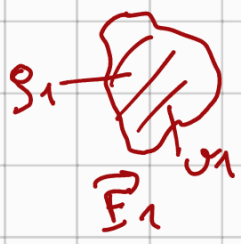
* zamana bağlı değil

ρ genelleştirebilirsek tümünü bir formülde alabiliriz (ilerde yapacağız)

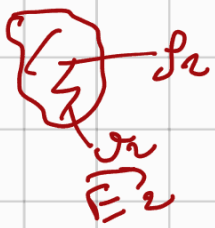
$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho}{r^3}\vec{r}\,d\tau$$

tüm uzayda

Sorduğum sorunun cevabı hocanın sorusu

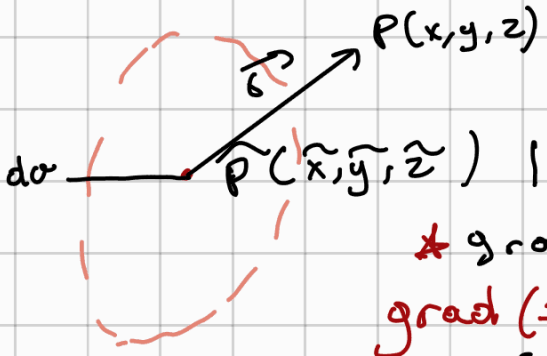


$\vec{E} = \vec{E}_1 + \vec{E}_2$ superposition



$\rho_1 + \rho_2 = \rho$ yazarsak genel formülü uygulayabiliriz

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$$\vec{r} = \overrightarrow{\tilde{P}P} = (x-\tilde{x})\vec{e}_x + (y-\tilde{y})\vec{e}_y + (z-\tilde{z})\vec{e}_z$$

$$|\vec{r}| = r = \sqrt{(x-\tilde{x})^2+(y-\tilde{y})^2+(z-\tilde{z})^2}$$

* gradiant x, y, z'ye göre

$$\text{grad}\left(\frac{1}{r}\right) = \text{grad}\left(r^{-1}\right)$$

$$= \text{grad}\left[\left[(x-\tilde{x})^2+(y-\tilde{y})^2+(z-\tilde{z})^2\right]^{-1/2}\right]$$

$$= -\frac{1}{2}\,2(x-\tilde{x})\left[(x-\tilde{x})^2+(y-\tilde{y})^2+(z-\tilde{z})^2\right]^{-3/2}\vec{e}_x$$

$$-\frac{1}{2}\,2(y-\tilde{y})\left[\text{-----}\right]^{-3/2}\vec{e}_y$$

$$-\frac{1}{2}\,2(z-\tilde{z})\left[\text{-----}\right]^{-3/2}\vec{e}_z$$

$$\text{grad}(1/r) = -\frac{1}{r^3}\left[(x-\tilde{x})\vec{e}_x + (y-\tilde{y})\vec{e}_y + (z-\tilde{z})\vec{e}_z\right)$$

$$= -\frac{\vec{r}}{r^3}$$

$$\boxed{\text{grad}\left(\frac{1}{r}\right) = \frac{-\vec{r}}{r^3}}$$

$\vec{E}$ yi böyle yazabiliriz

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\int \rho \left(-\frac{\vec{r}}{r^3}\right) dv \qquad \vec{r}: \text{grad}\left(\frac{1}{r}\right)$$

ρ sabit sayarsak

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\int_{\tilde{x}}\int_{\tilde{y}}\int_{\tilde{z}} \rho(\tilde{x},\tilde{y},\tilde{z})\ \text{grad}\left(1/r\right) d\tilde{x}\,d\tilde{y}\,d\tilde{z}$$

$$\star\ \vec{E} = -\text{grad}\left[\underbrace{\frac{1}{4\pi\varepsilon_0}\int_{\tilde{x}}\int_{\tilde{y}}\int_{\tilde{z}} \frac{\rho(\tilde{x},\tilde{y},\tilde{z})}{r}\, d\tilde{x}\,d\tilde{y}\,d\tilde{z}}\right]$$

$\Downarrow$

skaler potansiyel fonksiyonu

$$\star\ V(x,y,z) = \frac{1}{4\pi\varepsilon_0}\int_x\int_y\int_z \frac{\rho(\tilde{x},\tilde{y},\tilde{z})}{r}\, d\tilde{x}\,d\tilde{y}\,d\tilde{z} + V_0 \quad (V_0 \to \text{sabit})$$

sonsuza giderken $V_0 = 0$'dır

$$\star\ \boxed{\vec{E}(x,y,z) = -\text{grad}\, V(x,y,z)}$$

Noktasal için bakarsak

$$\vec{E} = -\text{grad}\, V = -\text{grad}\left[\frac{q}{4\pi\varepsilon_0}\cdot\frac{1}{r}\right]$$

$$= -\frac{q}{4\pi\varepsilon_0}\underbrace{\text{grad}\left(\frac{1}{r}\right)}_{-\frac{\vec{r}}{r^3}} = \frac{q}{4\pi\varepsilon_0}\frac{\vec{r}}{r^3}$$

$$\boxed{\underbrace{\text{rot}\,\vec{E}}_{0} = \underbrace{-\text{rot}\,\text{grad}\, V}_{=0} = 0}$$

$$\vec{\nabla} = \frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z$$

$$\Delta V = \text{div}\,\text{grad}\, V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad (\rho = 0 \text{ için dışında})$$

(V: skaler)

Laplacian (Laplasien)

$$\Delta \triangleq \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \triangleq \vec{\nabla}\cdot\vec{\nabla} = |\vec{\nabla}|^2$$

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\Delta \cdot \vec{A} = \frac{\partial^2}{\partial x^2}\vec{A} + \frac{\partial^2 \vec{A}}{\partial y^2} + \frac{\partial^2 \vec{A}}{\partial z^2}$$

$$\text{rot}\,\text{grad}\, f = \vec{\nabla}\times(\vec{\nabla} f) = (\vec{\nabla}\times\vec{\nabla}) f = 0$$

(f: skaler)

$$\vec{\nabla} \times \vec{\nabla} = \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right) \times \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right)$$

$\vec{e}_x \times \vec{e}_y = \vec{e}_z$
$\vec{e}_x \times \vec{e}_z = -\vec{e}_y$
$\vec{e}_y \times \vec{e}_z = \vec{e}_x$

$$= \frac{\partial^2}{\partial x^2}(\vec{e}_x \times \vec{e}_x) + \frac{\partial^2}{\partial x \partial y}\vec{e}_z - \frac{\partial^2}{\partial x \partial y} + \frac{\partial^2}{\partial y \partial z}\vec{e}_x$$

$$+ \frac{\partial^2}{\partial x \partial z}\vec{e}_y - \frac{\partial^2}{\partial y \partial z}\vec{e}_x$$

$$= 0$$

$$* \quad \boxed{\text{rot}\,\vec{E} = -\text{rot}\,\text{grad}\,V = 0}$$

Örnek

$\vec{E} = 3xy\vec{e}_y$ sağlayan yükleri bulun

$$\text{rot}\,\vec{E} = \vec{\nabla} \times \vec{E} = \left(\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z\right) \times (3xy\vec{e}_y)$$

$$= \frac{\partial}{\partial x}(3xy)\vec{e}_z - \frac{\partial(3xy)}{\partial z}\vec{e}_x$$

$$= 3y\vec{e}_z + 0 = 3y\vec{e}_z \neq 0$$

o zaman bu elektrostatikalan değildir
Böyle bir yük dağılımı yoktur.

$$\boxed{\text{Div rot}\,\vec{A} = 0} \quad \text{(vektör)}$$
her zaman geçerlidir

$$= \vec{\nabla}(\vec{\nabla} \times \vec{A})$$

burdan çıkan vektör $\vec{\nabla}$ ve $\vec{A}$ ya diktir

dikler skaler çarpımı 0 dır

$$\boxed{\text{rot rot}\,\vec{A} = \text{grad div}\,\vec{A} - \Delta\vec{A}}$$